

NAME

CLASS

ST ANDREW'S JUNIOR COLLEGE



JC 2 PRELIMINARY EXAMINATIONS

Chemistry

8873/01

Higher 1

17 Sept 2021

Paper 1

1400 – 1500

1 hour

Additional materials: Data Booklet and Optical Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and index number on the OAS provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OAS.

Read the instructions on the OAS very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of 15 pages.

- 1 The table shows the number of protons, neutrons and electrons present in some atoms and simple ions.

Which row describes two particles which are different isotopes of the same element with same charges?

	Particle 1			Particle 2		
	protons	neutrons	electrons	protons	neutrons	electrons
A	15	16	16	15	15	16
B	15	17	15	16	15	16
C	16	16	18	16	16	15
D	16	17	15	15	17	13

- 2 When 20 cm³ of a hydrocarbon was burnt in 150 cm³ of excess oxygen, the final gaseous mixture had a total volume of 110 cm³. On passing through aqueous potassium hydroxide, the volume of gaseous mixture decreased by 60 cm³. All gaseous volumes were measured under room temperature and pressure.

What is the formula of the hydrocarbon?

- A** C₃H₄
B C₃H₈
C C₆H₁₂
D C₆H₁₄
- 3 When 0.2 mol of potassium chlorate, KClO₃ is heated at its melting point, it forms potassium chlorate KClO₄ and potassium chloride.
- Which of the following statements is correct about this reaction?
- A** 0.2 mol of KClO₄ is formed.
B It is a disproportionation reaction.
C Water is formed as a by-product of this reaction.
D The oxidation state of Cl in KClO₃ is +7 while the oxidation of Cl in KClO₄ is +5.
- 4 Which one of the following ions will be deflected to the greatest extent in an electric field?
- A** ²³Na⁺
B ²⁷Al³⁺
C ³²S²⁻
D ¹⁴N³⁻

- 5 The first eight ionisation energies of an element **Q** (in kJ mol^{-1}) are as follows:

966, 1950, 2730, 4850, 6020, 12300, 15400, 18900

Which of the following statements about element **Q** is correct?

- A It has an even number of protons in its atoms.
 - B It can form a chloride with the formula QCl_5 .
 - C It can form a chloride that is trigonal planar in shape.
 - D It can form a basic oxide.
- 6 Why is the first ionisation of sulfur lower than that of phosphorus?
- A Sulfur is more electronegative than phosphorus.
 - B Sulfur forms anions more readily.
 - C The electron to be lost in sulfur is paired with another electron.
 - D The sulfur atom is larger than the phosphorus atom.
- 7 Which of the following descriptions of polarity and bond angle of the molecule is correct?

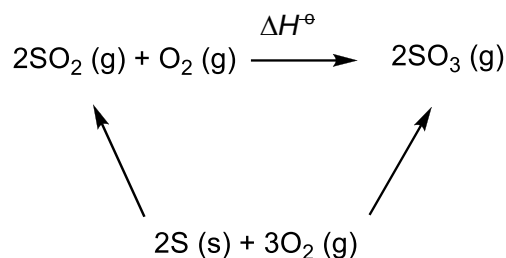
	Molecule	Polarity	Bond angle
1	BeCl_2	polar	104.5°
2	SiCl_4	non-polar	109.5°
3	XeCl_4	non-polar	90°

- A 1 and 2 only
 - B 2 and 3 only
 - C 2 only
 - D 3 only
- 8 Which of the following molecules required the highest pressure to be liquefied?
- A CH_4
 - B NH_3
 - C HF
 - D H_2O

- 9 When 25 cm³ of 2.0 mol dm⁻³ sodium hydroxide is completely neutralised by 25 cm³ of 3.0 mol dm⁻³ hydrochloric acid, the temperature of the mixture rose by 6.8 °C. Assuming negligible heat loss, what is the enthalpy of neutralisation in kJ mol⁻¹ for this reaction?

A + 14.2
 B + 18.9
 C - 14.2
 D - 28.4

- 10 Given the following enthalpy changes:



Using the energy cycle above, what is the enthalpy change of reaction, $\Delta H^\ominus$ in kJ mol⁻¹?

A +189
 B -687
 C -206
 D -1374

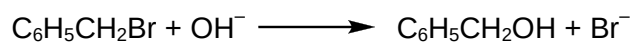
- 11 At 25°C, sucrose decomposes in acidic solution into glucose and fructose according to the rate equation below:

$$\text{Rate} = k [\text{sucrose}]$$

A sample of sucrose was allowed to decompose in acidic solution. At the end of 24 hours, it was found that only 12.5% of the original sample was left in the reaction flask. What is the half-life of the decomposition of sucrose in acidic solution?

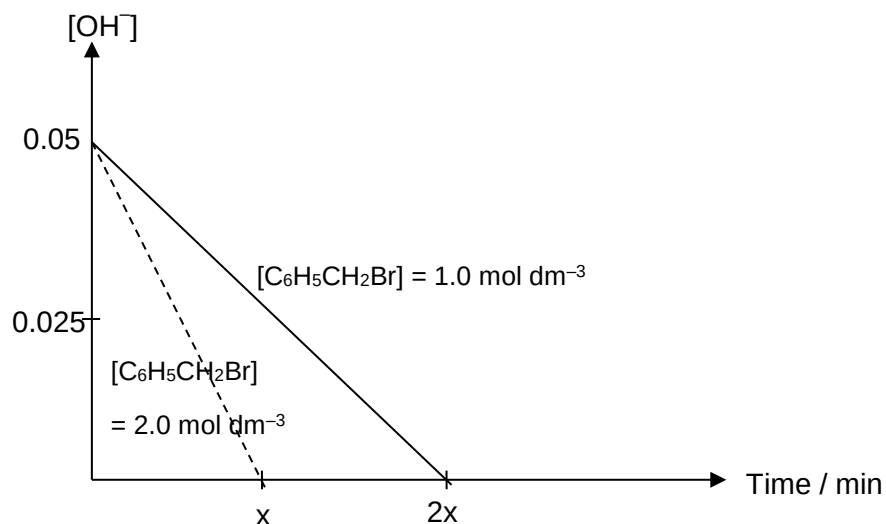
A 8 hours
 B 12 hours
 C 24 hours
 D 72 hours

12



The reaction kinetics of the above reaction is determined by monitoring the change in the concentration of OH^- with time.

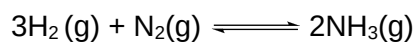
The results are shown below:



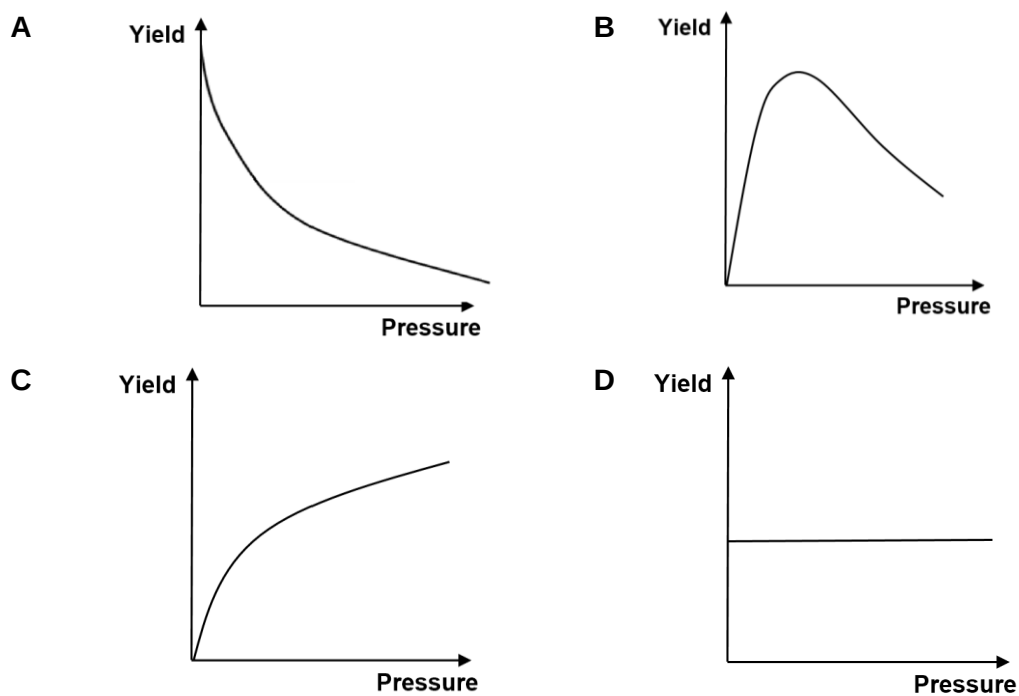
What is the rate equation for this reaction?

- A** rate = $k[\text{C}_6\text{H}_5\text{CH}_2\text{Br}]$
- B** rate = $k[\text{OH}^-]$
- C** rate = $k[\text{C}_6\text{H}_5\text{CH}_2\text{Br}]^2$
- D** rate = $k[\text{C}_6\text{H}_5\text{CH}_2\text{Br}][\text{OH}^-]$

- 13** Ammonia can be synthesised directly using a mixture of hydrogen gas and nitrogen gas in a reversible reaction as shown:



Which one of the following describes the change in yield of ammonia at equilibrium as pressure increases?

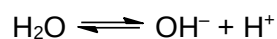


- 14** Which of the following mixtures are a *Bronsted-Lowry* acid and conjugate base pair?

- 1** H_2O and OH^-
- 2** H_2PO_4^- and HPO_4^{2-}
- 3** NaH and Na

- A** 1, 2 and 3 only
- B** 1 and 2 only
- C** 2 only
- D** 3 only

- 15** The value of the ionic product of water, K_w , varies with temperature.



temperature / °C	K_w / mol ² dm ⁻⁶
0	0.1×10^{-14}
10	0.3×10^{-14}
25	1.0×10^{-14}

Which row is correct?

The ionisation of water is

At 25 °C, water with a pH of 7.0 is

- | | | |
|----------|-------------|----------|
| A | endothermic | alkaline |
| B | endothermic | neutral |
| C | exothermic | alkaline |
| D | exothermic | neutral |

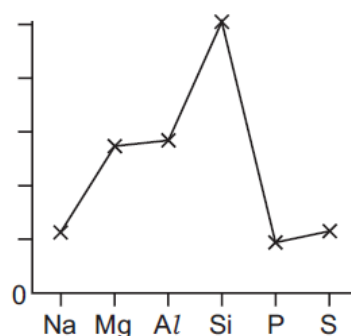
- 16** Solutions **E**, **F**, **G** and **H** contain either a strong or weak acid or base, but not necessarily in the same order. The concentration and pH for each solution are shown below.

Solution	Concentration / mol dm ⁻³	pH at 25°C
E	1.0	4.0
F	1.0	9.4
G	0.01	12.0
H	0.01	2.0

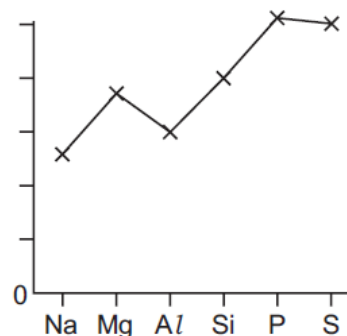
Which one of the following statements is a weak monobasic acid?

- | | |
|----------|----------|
| A | E |
| B | F |
| C | G |
| D | H |

- 17 The trends in three physical properties of the elements Na, Mg, Al, Si, P and S are shown in the following graphs.



Graph 1



Graph 2

Which of the following illustrate the correct physical property for the corresponding graphs above?

- | | Graph 1 | Graph 2 |
|---|-----------------------------------|-----------------------------------|
| A | melting point | electrical conductivity |
| B | 1 st ionisation energy | electrical conductivity |
| C | melting point | 1 st ionisation energy |
| D | 1 st ionisation energy | melting point |

- 18 Three samples of white powder, X, Y and Z were found. The possible identities of each sample are Al_2O_3 , P_4O_{10} and SiO_2 . Aqueous hydrochloric acid and sodium hydroxide were added to test their identities.

	HCl (aq)	NaOH (aq)
X	No reaction	Reaction
Y	Reaction	Reaction
Z	No reaction	No reaction

Which of the following shows the correct identities of X, Y and Z?

- | | X | Y | Z |
|---|---------------------------|---------------------------|---------------------------|
| A | Al_2O_3 | P_4O_{10} | SiO_2 |
| B | P_4O_{10} | SiO_2 | Al_2O_3 |
| C | P_4O_{10} | Al_2O_3 | SiO_2 |
| D | SiO_2 | Al_2O_3 | P_4O_{10} |

19 Which of the following statements about the trends in the properties of halogens are correct on descending the group?

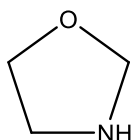
- 1 The electronegativity decreases.
- 2 The volatility increases.
- 3 Their reactivity as oxidising agents decreases.

- A** 1, 2 and 3 only
B 2 and 3 only
C 1 and 3 only
D 2 only

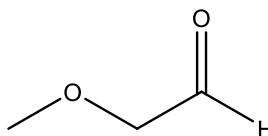
20 How many tertiary alcohols have the molecular formula $C_6H_{14}O$?

- A** 1
B 2
C 3
D 4

21 Which one of the following statements best accounts for the difference in the boiling points of oxazolidine and compound **W**?



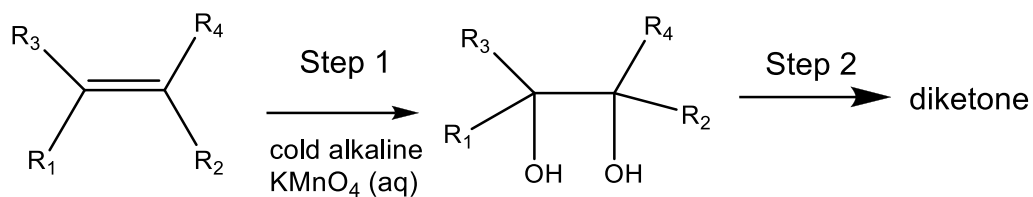
Oxazolidine



Compound **W**

- A** Oxygen is more electronegative than nitrogen.
B Oxazolidine has a smaller surface area than **W** for intermolecular interactions.
C Oxazolidine has 3 lone pairs of electrons while **W** has 4 lone pairs of electrons.
D Hydrogen bonding present between oxazolidine molecules is stronger than the permanent dipole - permanent dipole interactions present between **W** molecules.

The following reaction scheme occurs in an alkene. It is relevant for Question **22** and **23**.



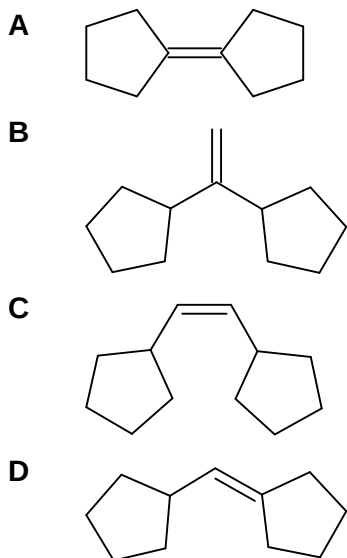
J

R_1 , R_2 , R_3 and R_4 represent alkyl groups or H.

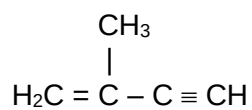
22 What types of reaction occur in Step 1 and Step 2?

	Step 1	Step 2
A	addition	addition
B	oxidation	oxidation
C	substitution	addition
D	oxidation	reduction

23 What is a possible identity of the starting material, **J** to give a diketone?

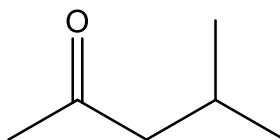


- 24 Which of the following options shows the correct number and type of bonds in Molecule **K**?

Molecule **K**

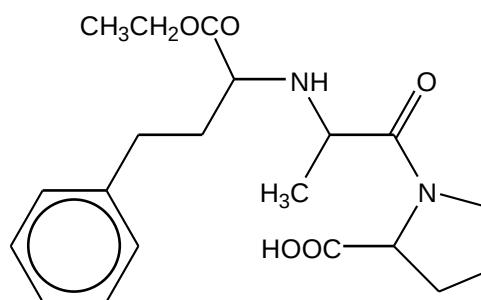
	Number of σ - bonds	Number of π - bonds
A	4	3
B	8	5
C	10	2
D	10	3

- 25 How many different mono-brominated compounds can be formed when Compound **H** reacts with gaseous bromine under the presence of UV light?

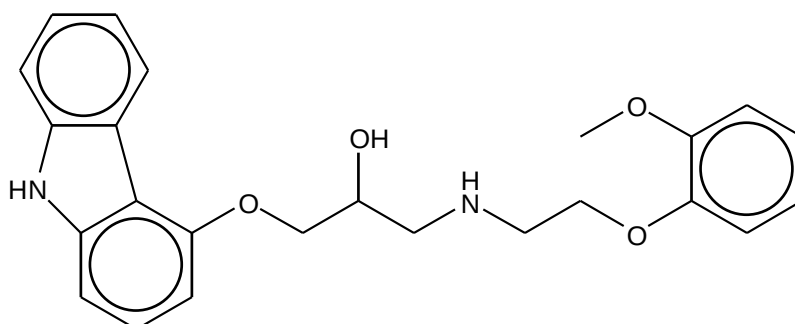
Compound **H**

- | | |
|----------|---|
| A | 2 |
| B | 3 |
| C | 4 |
| D | 5 |

- 26** Enalapril and Carvedilol are two compounds that are used in the treatment of heart failure and disease.



Enalapril



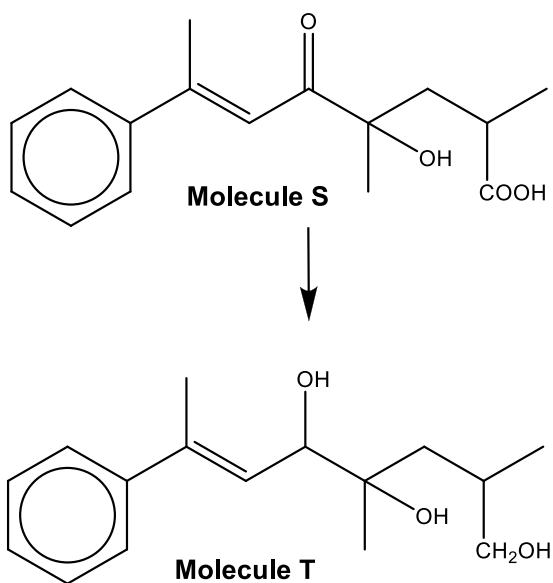
Carvedilol

Which of the following can be used to distinguish between the two compounds?

- 1 Na_2CO_3 (aq)
- 2 H_2/Pt
- 3 Ethanolic KOH

- A** 1 only
B 1 and 2 only
C 2 and 3 only
D 1 and 3 only

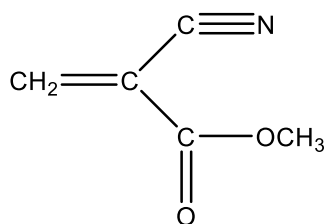
27 Which of the following reducing agents would convert Molecule **S** to Molecule **T**?



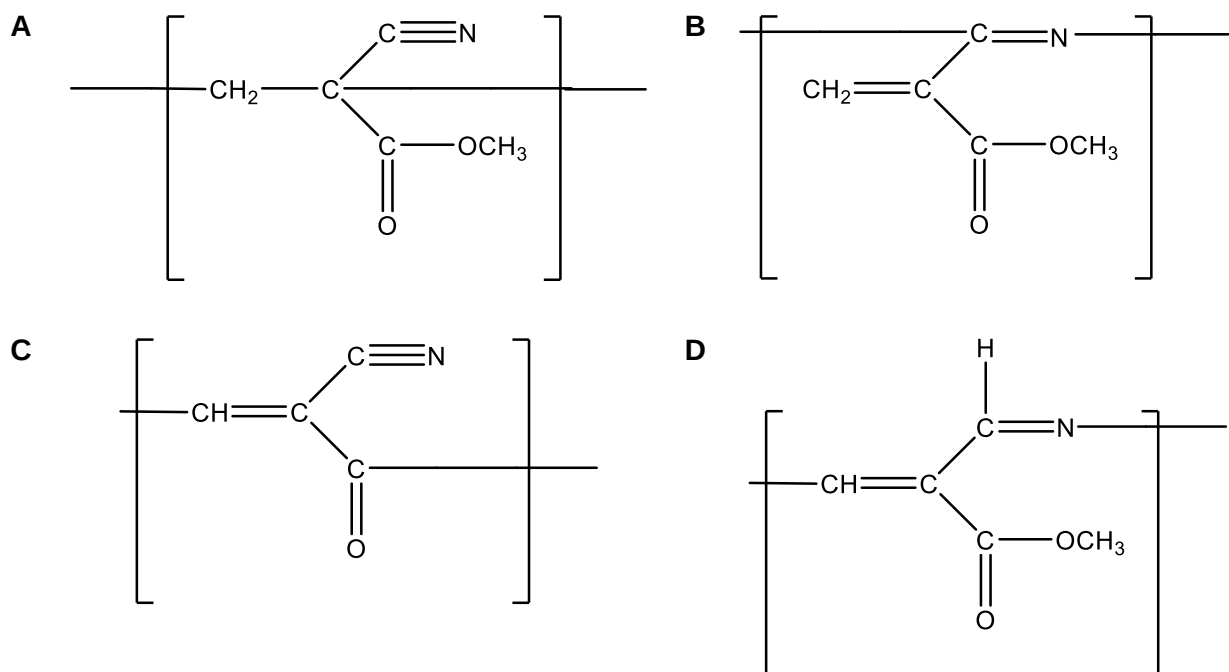
- 1 LiAlH_4 / dry ether
- 2 NaBH_4
- 3 H_2 / Ni

- A 1, 2 and 3 only
- B 1 and 2 only
- C 1 only
- D 2 only

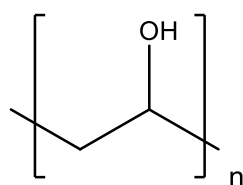
28 "Superglue" contains the following compound



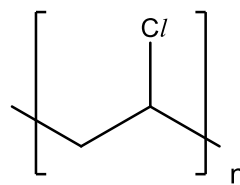
It is rapidly polymerised by traces of base on the surface of the objects to be stuck together. Which of the following represents the repeat unit of the polymerised form?



29 Which of the following about polyvinylalcohol (PVA) and polyvinylchloride (PVC) is correct?



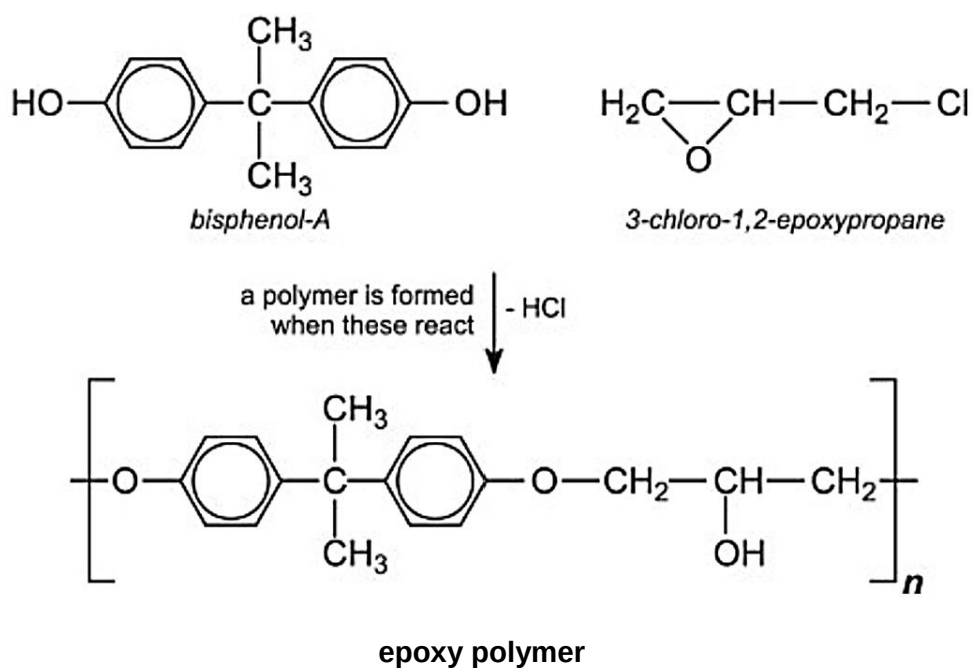
PVA



PVC

- A PVC is a poorer water resistant polymer than PVA.
- B PVC can react with NaOH via a substitution reaction
- C Both PVC and PVA has hydrogen bonding present between its polymeric chains.
- D PVA is used in eye drops because of the strong permanent dipole – permanent dipole interaction with water molecules.

30 Epoxy polymers can be made as shown:



Source: <https://www.essentialchemicalindustry.org/materials-and-applications/paints.html>

Which of the following statements is most likely to be **false** about epoxy polymer?

- A** It is a condensation polymer.
- B** It does not form cross-links with its own polymeric chains.
- C** It is water-soluble.
- D** It does not degrade via hydrolysis.

~ END OF PAPER ~